

Aditya Shirsatrao

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Solapur, Maharashtra, India

Education

N. K. Orchid College of Engineering & Technology Solapur, Maharashtra
Bachelor of Technology in Artificial Intelligence and Data Science – CGPA: 8.5/10 Aug. 2023 – Present (Expected 2027)
Coursework: Data Structures & Algorithms, Operating Systems, Database Management Systems, Distributed Systems, Machine Learning, Neural Networks, OOP.

Experience

Fullstack Web Development Intern Unified Mentor Pvt. Ltd
Backend, Distributed Systems & AI-Integrated Products Jul. 2025 – Jan. 2026
• Architected an AI-powered code review platform processing GitHub pull requests asynchronously; integrated Kafka message queues and Redis caching, reducing analysis latency by 45% through parallel distributed workers and AST-based code search.
• Engineered a real-time collaborative IDE with sub-100ms sync for concurrent users using WebSockets, Socket.IO, Redis pub/sub, and CRDT-based operational transforms; integrated Monaco Editor with live multiplayer cursors.
• Built a distributed ML inference platform with autoscaling endpoints, model versioning, and Prometheus monitoring; improved throughput by 60% under concurrent load via request batching and inference caching.
AI/ML Lead Google Developer Groups on Campus (GDG-NKOCET)
Community Leadership & Mentorship Sep. 2025 – Present
• Lead AI/ML initiatives and technical workshops for a 200+ student community; mentored juniors on system design, Git/GitHub workflows, and production ML engineering.

Projects

Fraud Detection System | FastAPI, Autoencoders, React, D3.js | Code Dec 2024 – Present
• Architected a real-time anomaly detection engine using unsupervised Autoencoders, identifying novel fraud patterns across 500k+ transactions without prior labeling.
• Optimized inference to <50ms by wrapping the scikit-learn pipeline in a FastAPI microservice with async I/O, supporting 100 requests/second.
Privacy-First LLM Assistant | RAG, Llama 3, ChromaDB, Ollama | Code Nov 2024 – Dec 2024
• Engineered a local Retrieval-Augmented Generation system for secure PDF/TXT querying, ensuring zero external data exfiltration.
• Implemented Hybrid Search (dense vector + sparse keyword) with cross-encoder re-ranking, improving retrieval precision by 40% on technical corpora.
• Designed a persistent vector store with ChromaDB, enabling incremental indexing of 100+ documents without full rebuilds.

Open Source

- Merged 50+ pull requests into high-profile open source repos including Google (OSS-Fuzz), Microsoft (VibeVoice), Apache (DataFusion), and Hedera Hashgraph (Hiero SDK).
- Hiero SDK (Hedera): Refactored the EVM transaction builder to support direct alias conversion, cutting boilerplate by 30% and eliminating type redundancy errors (PR #1030).

Technical Skills

Languages: Python, Java, C++, JavaScript (ES6+), TypeScript, SQL
Frameworks: FastAPI, Node.js, Express.js, React.js, Next.js, Flask
Infrastructure: Docker, Kubernetes, Redis, Kafka, GitHub Actions, AWS (EC2/S3/Lambda), Linux
Core Concepts: DSA (366+ LeetCode), Distributed Systems, Microservices, CI/CD, Asynchronous Programming, RESTful APIs

Achievements & Certifications

Competitive Programmer: 366+ LeetCode problems solved (104 Easy, 176 Medium, 86 Hard) + 50 SQL problems.
IEEE First-Author Publication presented at ICCTWC 2026; 1st Place at Hack to Future 3.0 (600+ teams) and 1st Runner-Up at Orchathon 2K26 (800+ teams).
Certifications: JPMorgan Chase & Electronic Arts Software Engineering Virtual Experience; NPTEL Programming in Java (Elite+Gold, 91/100).